COMP3055
Machine Learning

Topic 3 – Data Collection

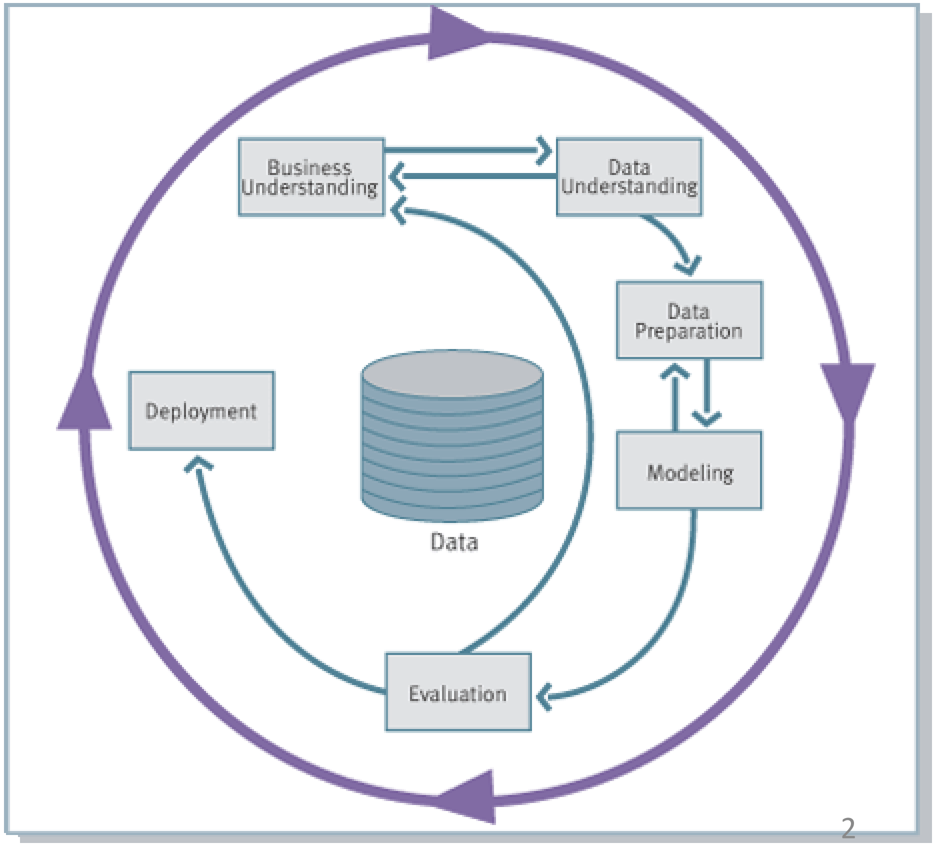
2025 Autumn

Data Mining Process Model

CRoss Industry Standard Process for Data Mining (**CRISP-DM**)

Industry Standard

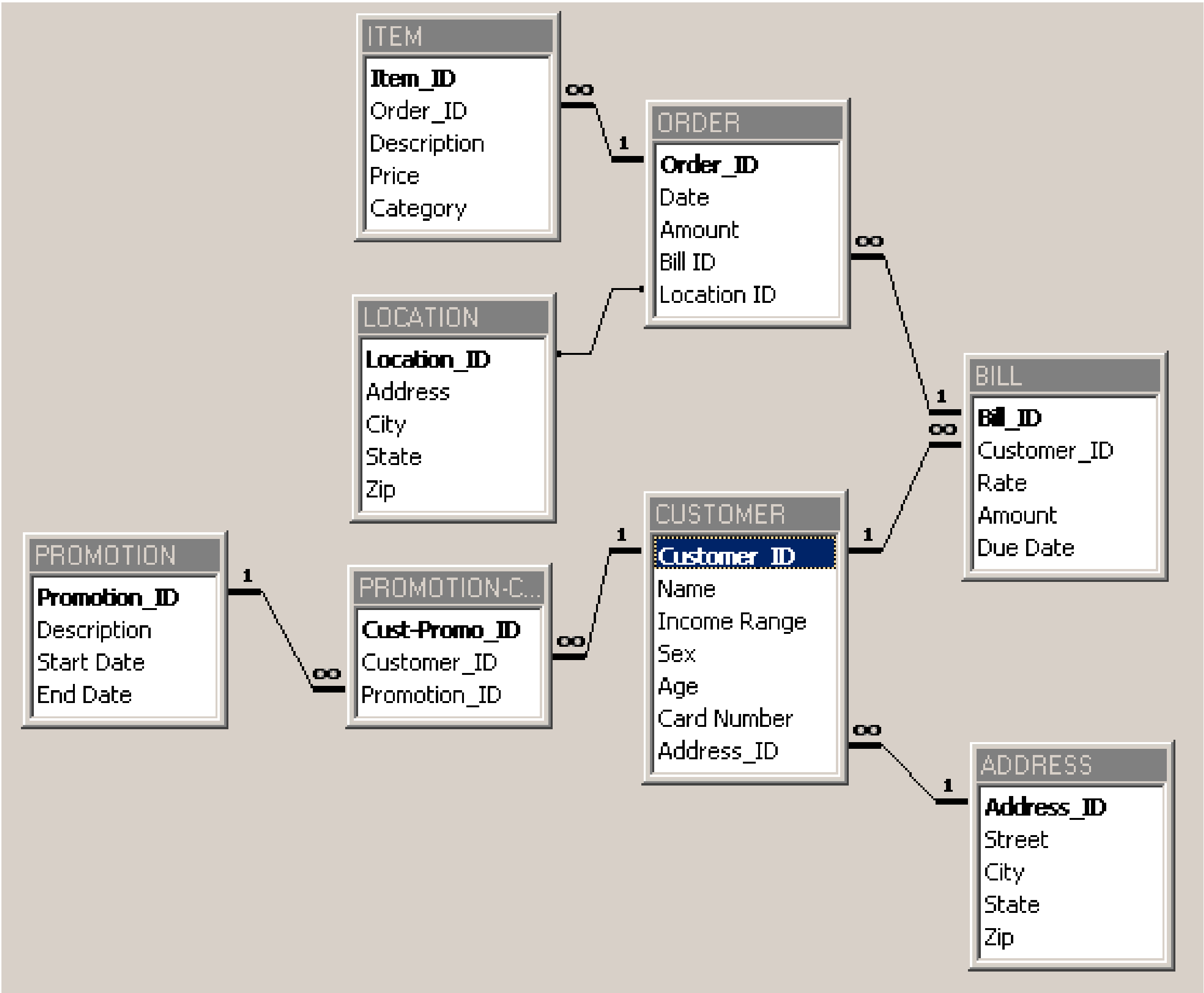
The most popular data mining methodology model according to KDNuggets.com



Step 1: Business Understanding

- Define the problem
- Choose a machine learning model(s)
- Estimate project cost
- Estimate project completion time
- Address legal issues
- Develop a maintenance plan

Step 2: Data Understanding



Step 3: Data Preparation

噪声处理：去重、纠错、平滑（如移动平均/分箱）。

缺失处理：删行、均值/众数填补、**相似样本**插补（如KNN）。

变换：归一化/标准化、类型转换、特征与样本选择。实战中，这一步最花时间

- Noisy data
  – Locate duplicate records
  – Locate incorrect attribute values
  – Smooth data
- Missing data
  – Discard records with missing values
  – Replace missing real-valued items with the class mean
  – Replace missing values with values found within *highly similar* instances
- Data transformation
  – Data normalization
  – Data type conversion
  – Attribute and instance selection

Step 4: Modeling

- Choose training and test data
- Designate a set of input attributes
- If learning is supervised, choose one or more output attributes
- Select learning parameter values
- Train the model

Step 5: Evaluation

- Statistical analysis
- Heuristic analysis
- Experimental analysis
- Human analysis

Measures of Effectiveness of the Model

- Accuracy

  Percentage of total predictions that were correct
- Return on investment

  Cost-benefit ratios
- Explanation

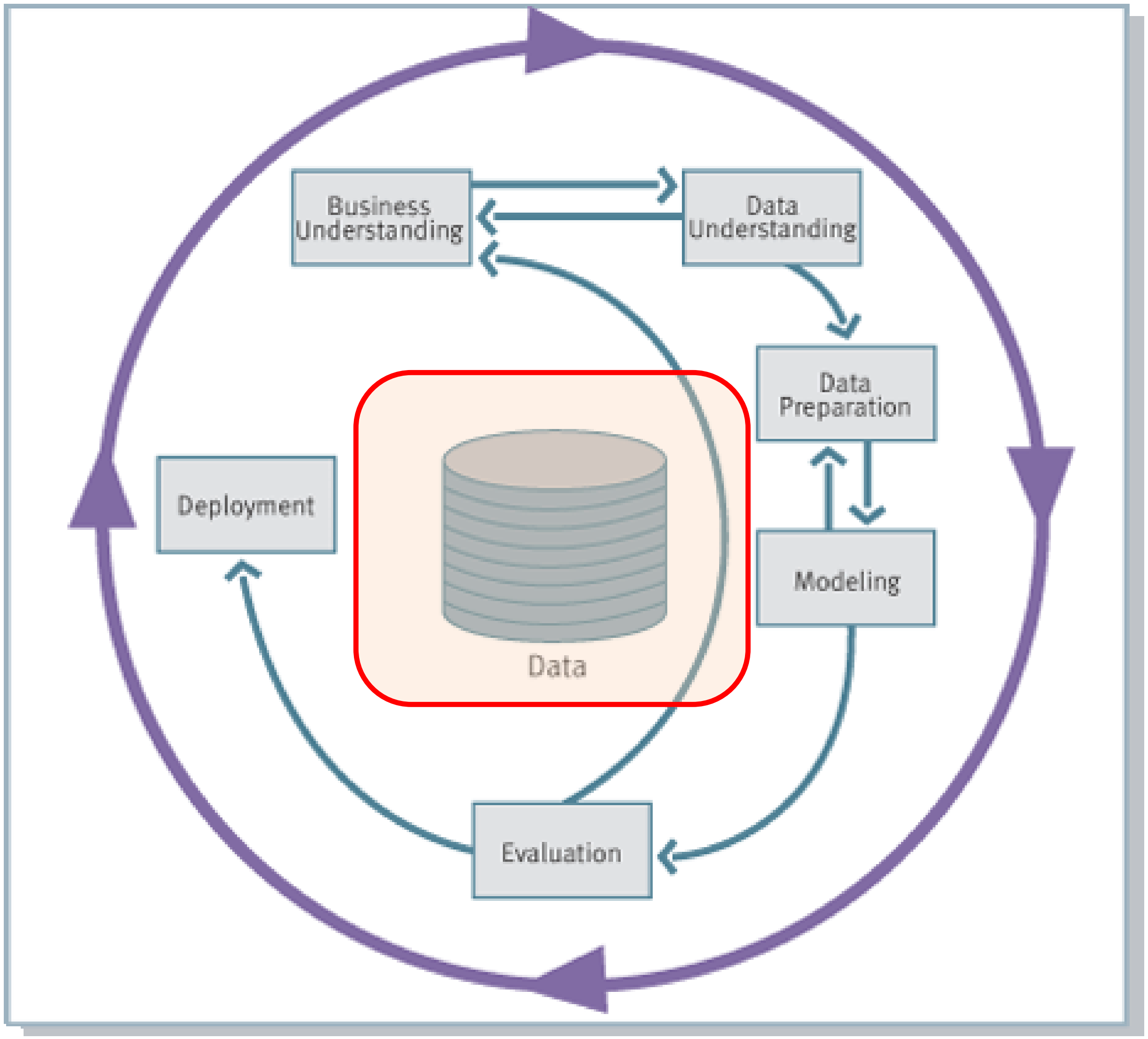
  Able to justify intuition
- Validation

  Automated checking of correctness, indexes

Step 6: Deployment 部署

- Apply the model to real world usage
- Apps, API, etc.
- Regularly update the model with new data
- …

You Need Collect Data before Learning Starts!



What is Data?

数据是一组数据对象及其相关属性的集合。
换句话说，一份数据集通常是一张表：
每一行（row） = 一个对象（object）
每一列（column） = 一个属性（attribute）

- **Collection of data objects** and their associated attributes
- An **attribute** is a property or characteristic of an object 属性是对象的特征或性质（property or characteristic）。
  - Examples: eye color of a person, temperature, etc.
  - Attribute is also known as variable, field, characteristic, or feature
- A **collection of attributes** describe an object
  - Object is also known as record, point, case, sample, entity, or instance

Attributes

Objects

Tid	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Types of Attributes

There are different types of attributes

- Nominal 名义
  - Examples: ID numbers, eye color, zip codes
- Ordinal 有序
  - Examples: rankings (e.g., taste of potato chips on a scale from 1-10), grades, height in {tall, medium, short}
- Interval 区间
  - Examples: calendar dates, temperatures in Celsius or Fahrenheit.
- Ratio 比率
  - Examples: temperature in Kelvin, length, time, counts

Attribute Type	Description	Examples	Operations
Nominal	The values of a nominal attribute are just different names, i.e., nominal attributes provide only enough information to distinguish one object from another.	zip codes, employee ID numbers, eye color, sex: {*male*, *female*}	mode, entropy, contingency correlation, χ^2 test
Ordinal	The values of an ordinal attribute provide enough information to order objects.	hardness of minerals, {*good, better, best*}, grades, street numbers	median, percentiles, rank correlation, run tests, sign tests
Interval	For interval attributes, the differences between values are meaningful, i.e., a unit of measurement exists.	calendar dates, temperature in Celsius or Fahrenheit	mean, standard deviation, Pearson's correlation, t and F tests
Ratio	For ratio variables, both differences and ratios are meaningful.	temperature in Kelvin, monetary quantities, counts, age, mass, length, electrical current	geometric mean, harmonic mean, percent variation

Discrete and Continuous Attributes

Discrete attribute 离散属性

拥有**有限（finite）或可数（countable）**的取值集合。
每个属性值之间是**可数的间隔**，即值与值之间有“ 间断 ”。

- Has only a **finite** or countable set of values.
- Examples: zip codes, number of employees, or sale counts.
- Often represented as **integer** variables.
- Note: binary attributes are a special case of discrete attributes.

通常用**整数（integer）**表示；
如果属性只能取 0 或 1，就叫做**二元属性（binary attribute）**；二元属性是离散属性的一种特殊情况。

Continuous attribute 连续属性

- Has **real** numbers as attribute values.

取值在一个**连续区间**中，可以是任意实数（real numbers）。值与值之间**没有间断**，可以无限细分

- Examples: temperature, stock prices, or net income.
- Continuous attributes are typically represented as **floating-point** variables.

一般用**浮点数（floating-point）**表示；常见于物理量、金融量、时间量等连续变化的概念

Types of Data

Record 记录型

- Data matrix
- Transaction data

Graph 图结构型

- Social networks
- Molecular structures 分子结构

Ordered 有序型

- Spatial data 空间数据
- Temporal data 时间数据
- Sequential data 序列数据
- Genetic sequence data 基因序列数据

We often deal with a mixture of different types data

Record Data

- Data that consists of a collection of records, each of which consists of a fixed set of attributes.

记录型数据是由**多条记录（records）组成的集合，每条记录包含一组固定的属性（attributes）**。
换句话说：
每一行（row） = 一条记录（record）
每一列（column） = 一个属性（attribute）

Tid	**Refund**	**Marital Status**	**Taxable Income**	**Cheat**
1	Yes	Single	125K	**No**
2	No	Married	100K	**No**
3	No	Single	70K	**No**
4	Yes	Married	120K	**No**
5	No	Divorced	95K	**Yes**
6	No	Married	60K	**No**
7	Yes	Divorced	220K	**No**
8	No	Single	85K	**Yes**
9	No	Married	75K	**No**
10	No	Single	90K	**Yes**

Data Matrix 数据矩阵

- If data objects have the same fixed set of **numeric** attributes, the data objects can be thought of as points in a multi-dimensional space, where each dimension represents a distinct attribute.

- Such data set can be represented as an `m` by `n` matrix, where there are `m` rows (one for each object) and `n` columns (one for each attribute).

Projection of x Load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	2.7	1.2
12.65	6.25	16.22	2.2	1.1

Transaction Data 是一种特殊的记录型数据（Record Data），其中每条记录（transaction）由一组**项目（items）**组成。
简而言之：
一条记录 = 一次交易（Transaction）
一次交易包含若干“ 商品 ”或“ 事件 ”
每个商品或事件称为一个 Item（项）

Transaction Data

- A special type of record data, where
  - Each record (transaction) involves a set of items.
  - For example, consider a grocery store. The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items.

TID	*Items*
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Graph Data

Graph Data 是一种以节点（Nodes/Vertices）和边（Edges/Links）为基础的数据类型，主要用于表示对象之间的关系或连接。

简单理解：

节点（Node）　表示“对象”或“实体”

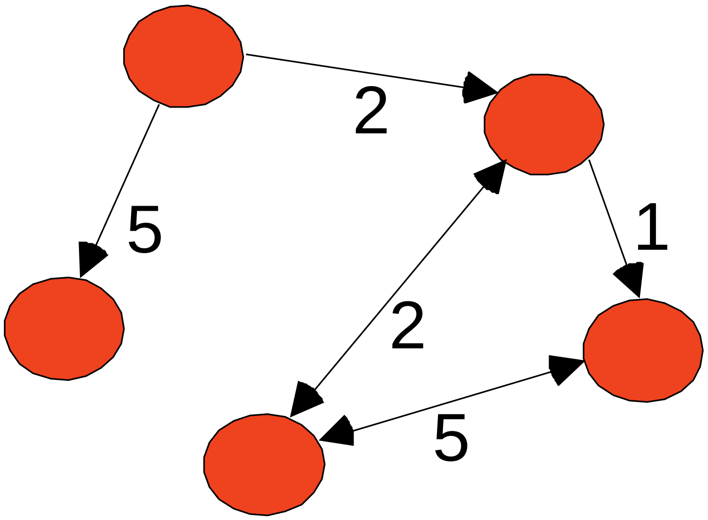
边（Edge）　表示“关系”或“交互”

权重（Weight（可选）　表示关系的强度、频率或距离等。

Examples:

– Representation of HTML Links

– Social Networks



Ordered Data

Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
GCCAAGTAGAACACGCGAAGCGC
TGGGCTGCCTGCTGCGACCAGGG
```

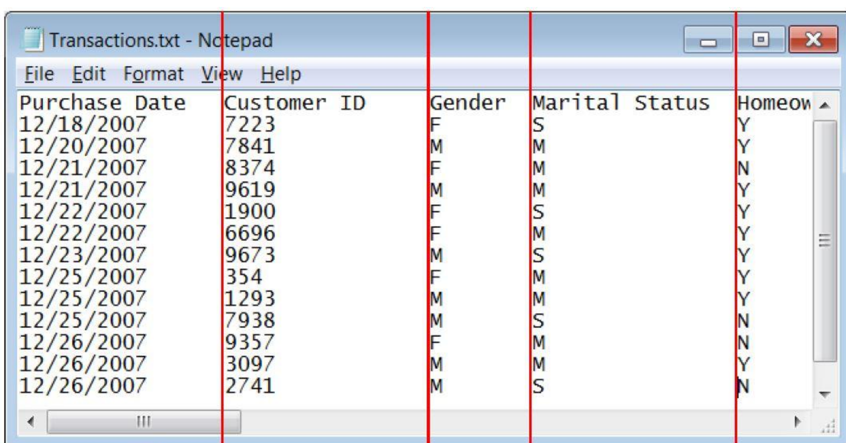
Obtaining Data

- **Data sources**
  - Obtained directly from owner: text file or relational database.
  - Collect public available data: web.
- **Unstructured VS. structured**
  - Text file: can be unstructured or structured.
  - Relational database: structured.
  - Web: unstructured.
  - Need to structure the data if necessary.
- **Data need to be cleaned**
- **Tools for data collection and cleaning**
  - Python, R, Excel, SQL, etc.

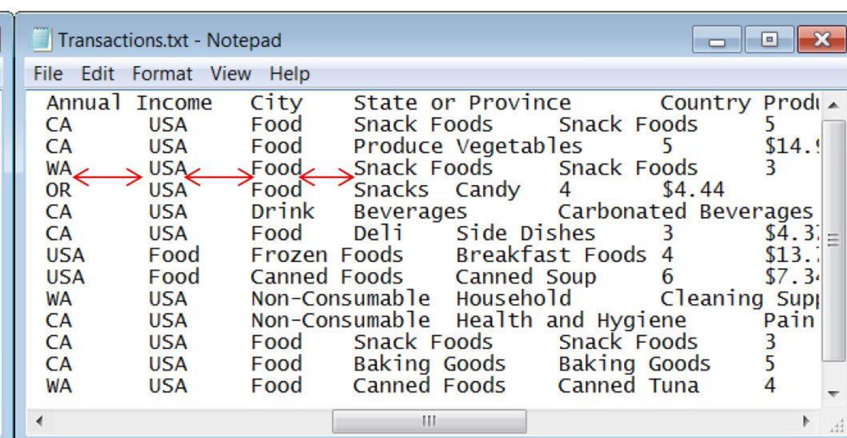
Text Files

- Most companies use proprietary software to store data that can be exported into text files.
  - Usually with .txt extension.
  - Sometimes with .csv extensi
- Commonly used text file format
  - Fixed-width: each attribute value starts and stops a fixed positions (columns) in the lines.
  - Delimited: there is a delimiter character, usually a tab, space, or **comma**, that separates different values in the lines.
  - **CSV**: comma separated values

格式类型	定义	特征
Fixed-width（定宽格式）	每个属性值在固定列宽范围内出现。即每列的起始与终止位置固定。	格式整齐，易读，但维护成本高。
Delimited（分隔符格式）	用特定字符（称为 *delimiter*）分隔各个属性值。	灵活、节省空间，是最常用格式。
CSV（Comma-Separated Values）	一种特殊的分隔符格式，使用逗号 , 分隔值。	广泛用于 Excel、Python、R。



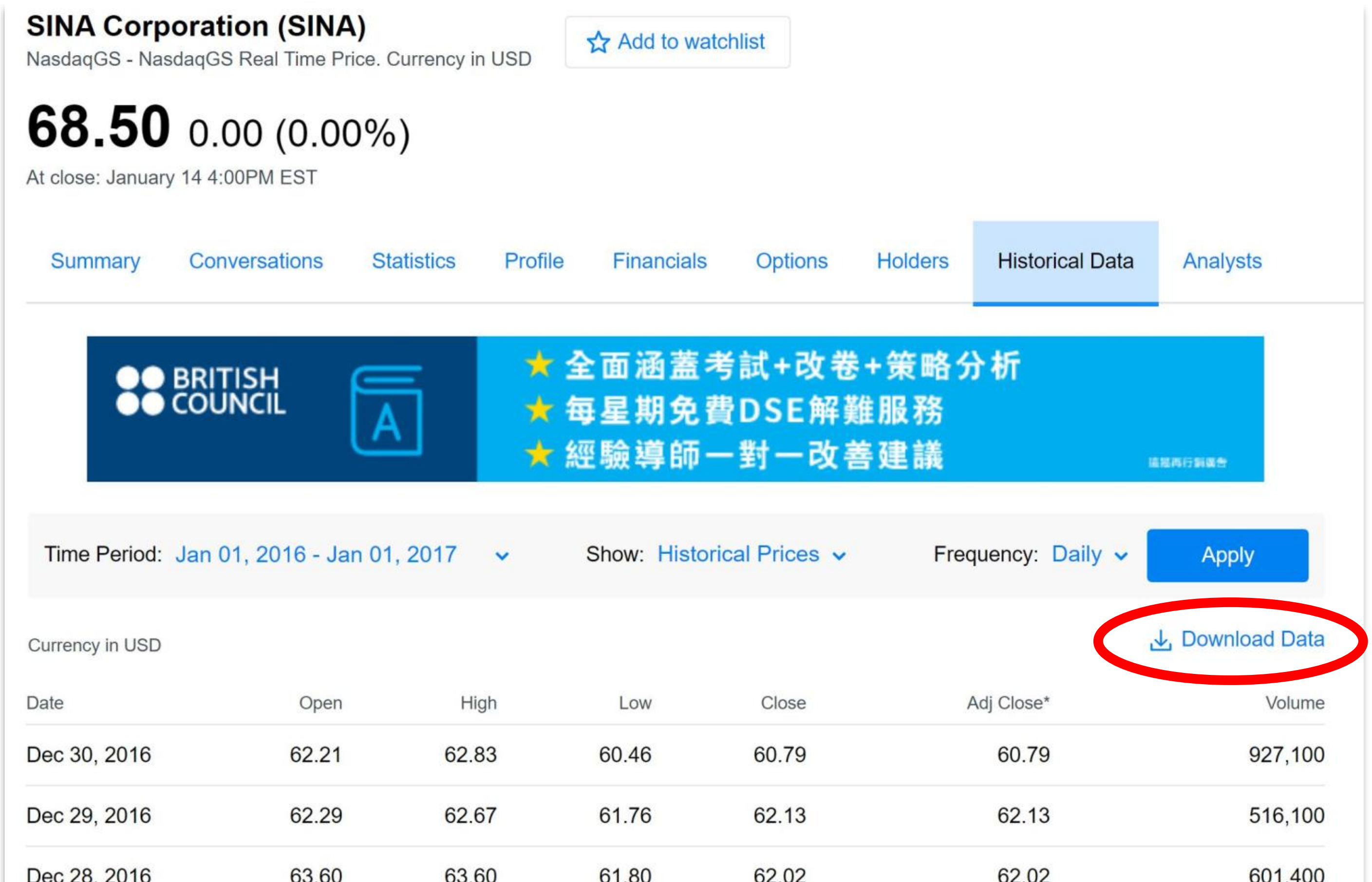

```
Purchase Date   Customer ID   Gender   Marital Status   Homeow
12/18/2007      7223          F        S                Y
12/20/2007      7841          M        M                Y
12/21/2007      8374          F        M                N
12/21/2007      9619          M        M                Y
12/22/2007      1900          F        S                Y
12/22/2007      6696          F        M                Y
12/23/2007      9673          M        S                Y
12/25/2007      354           F        M                Y
12/25/2007      1293          M        M                Y
12/25/2007      7938          M        S                N
12/26/2007      9357          F        M                N
12/26/2007      3097          M        M                Y
12/26/2007      2741          M        S                N
```




```
 Annual Income   City    State or Province     Country Produ
 CA     USA    Food    Snack Foods     Snack Foods      5
 CA     USA    Food    Produce Vegetables      5        $14.9
 WA     USA    Food    Snack Foods     Snack Foods      3
 OR     USA    Food    Snacks   Candy  4       $4.44
 CA     USA    Drink   Beverages       Carbonated Beverages
 CA     USA    Food    Deli     Side Dishes    3        $4.3
 USA    Food   Frozen Foods     Breakfast Foods 4       $13.
 USA    Food   Canned Foods     Canned Soup    6        $7.3
 WA     USA    Non-Consumable   Household      Cleaning Supp
 CA     USA    Non-Consumable   Health and Hygiene      Pain
 CA     USA    Food    Snack Foods     Snack Foods      3
 CA     USA    Food    Baking Goods    Baking Goods     5
 WA     USA    Food    Canned Foods    Canned Tuna      4
```

CSV Example

- Yahoo stock price historical data
  - Example: Sina stock price from Jan 1st, 2016 to Jan 1st 2017
  - http://finance.yahoo.com/quote/SINA/history?period1=1451577600&period2=1483200000&interval=1d&filter=history&frequency=1d



CSV Example

- Manual download from Yahoo
  - http://chart.finance.yahoo.com/table.csv?s=SINA&a=0&b=1&c=2016&d=0&e=1&f=2017&g=d&ignore=.csv
  - Save to local disk as csv file, e.g. “table.csv”
  - Use Python to read csv file
- Automatically download from Yahoo
  - Use Python for both downloading and reading

```
# Load Yahoo stock price for Sina
import numpy as np
# URL for SINA from Jan 1st 2016 to Jan 1st 2017
# url = "table.csv"
url = "http://chart.finance.yahoo.com/table.csv?s=SINA&a=0&b=1&c=2016&d=0&e=1&f=2017&g=d&ignore=.csv"
# load the CSV file as a numpy matrix
dataset = np.genfromtxt(url, dtype=None, skip_header=1, delimiter=",")
print(dataset[0])
```

Relational Database

关系型数据库是由**多个相互关联的表（tables）**组成的系统，每个表都是“行×列”的矩阵结构。

- A **relational database** is a set of related tables, where each table is a rectangular arrangement of fields and records.
  - Rows corresponding to records.
  - Columns corresponding to fields.
  - Primary key contains unique values to index data.
  - Foreign key links tables and can contain duplicate values.

组成部分	含义
Row（行）	表示一条完整的记录（record）
Column（列）	表示字段或属性（field/attribute）
Primary Key（主键）	每条记录的唯一标识符，不能重复
Foreign Key（外键）	用来连接不同表之间的字段

- Python allows you to import data from many database packages
  - You may need to install drive for different database, e.g. MySQLdb for MySQL database, before use.
- Basic steps
  - Make connection with host name, user name, password, database name, etc.
  - Create a cursor to the connected database
  - Execute SQL queries and fetch the data through the cursor
  - Close the database

Relational Database

```
import MySQLdb

db = MySQLdb.connect(host="localhost", # your host, usually localhost
user="john", # your username
passwd="megajonhy", # your password
db="jonhydb") # name of the data base

# you must create a Cursor object. It will let
# you execute all the queries you need
cur = db.cursor()

# Use all the SQL you like
cur.execute("SELECT * FROM YOUR_TABLE_NAME")
# print all the first cell of all the rows

for row in cur.fetchall():
    print row[0]

db.close()
```

Data from the Web

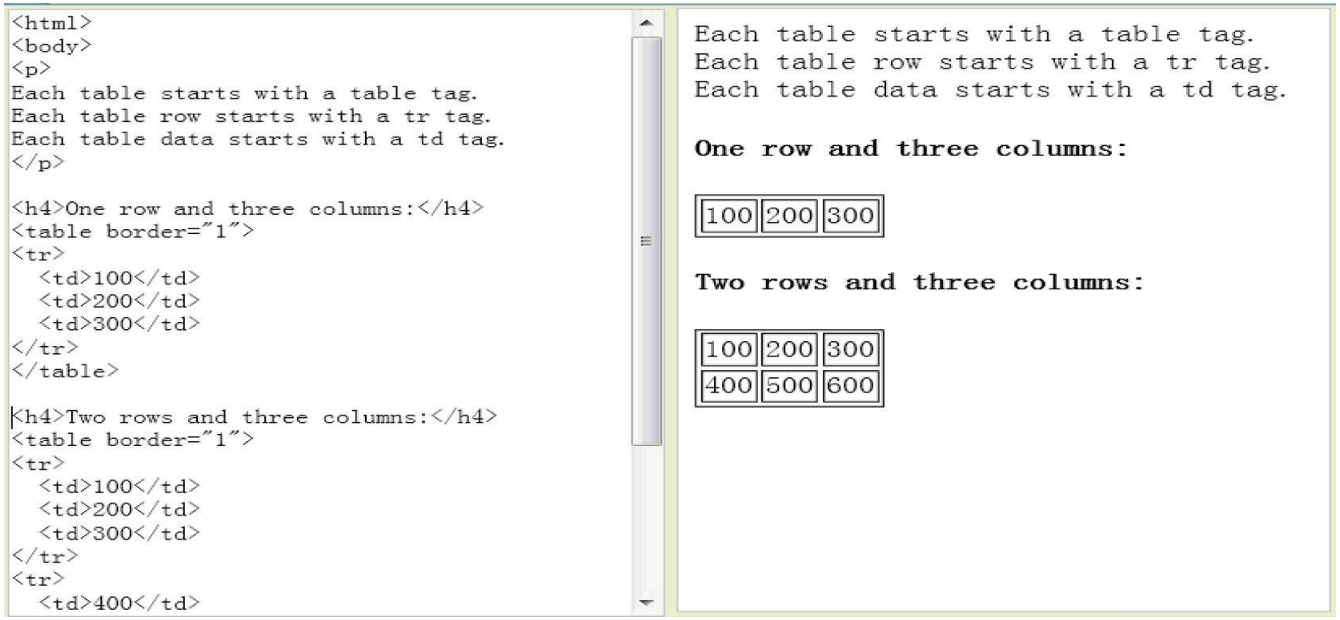
- Web sites containing data are structured in all sorts of ways, and the steps required to collect the data for analysis vary greatly.

- No matter the server side is a program or file, what the client side received (and the browser shows) are all files.
  - Usually in HTML (hypertext markup language) format.
  - HTML uses tags for displaying various items on the web page.

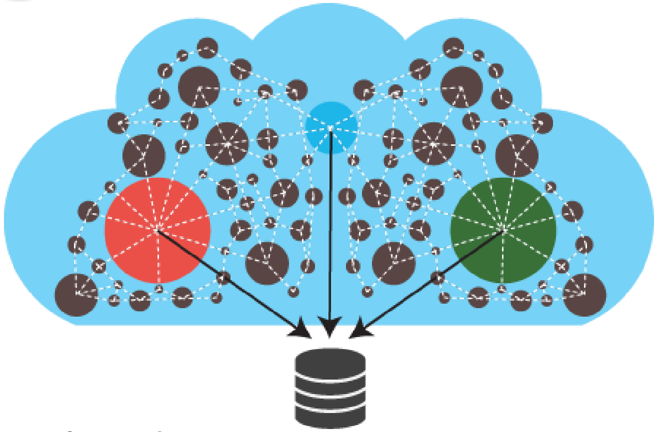
HTML File and Representation

大多数 HTML 标签是用来**控制网页样式和布局**的，而不是实际数据本身。
它们决定了网页的外观（例如字体、颜色、图片、按钮、对齐方式等），但并不一定包含分析所需的“有意义数据”。

- Most **html tags** control font, color, images, actions, etc., which are not related to the content of data.
- The layout of html is often controlled using table (old fashion) or CSS (Cascading Style Sheets, new fashion).
- Tablet data are normally put inside table tag, but many data we are interested are not in the table.



Web Crawling



- Find out the patterns of URL, e.g.
  - http://finance.yahoo.com/q?s=SINA
  - actual address: http://finance.yahoo.com/q
  - Parameter: **s=SINA**
  - **name=value** pairs separated by &
- Write a program to generate the URLs and download.
  - By changing parameters and name=value pairs
- Sometimes need to find and download the URLs inside a web page (spidering).
  - URLs normally start with "http" in raw html files.
- Parsing the web pages to collect useful data
  - **Identify** where the useful data is in a web page.
  - Finding **patterns** to help the program auto locate the data by looking at raw html files.
  - Use Python Regular Expression to locate the data according to the pattern.

Unstructured Data Parsing

Regular Expression

- **Python**, Java, etc.
- Find patterns from the “unstructured” data, including text file, HTML file, or other files.
- Define such patterns using regular expression grammar.
- Read files into a string. The processing engine will extract data from the string that meets this pattern.
- More on this using Python later!

```
import urllib.request
import re

url="http://google.com"

# regular expression for locating title
these_regex=b"<title>(.+?)</title>“
pattern=re.compile(these_regex)

# load the url
with urllib.request.urlopen(url) as response:
    html = response.read()

# find the pattern in the downloaded file
titles=re.findall(pattern, html)
print(titles)
```

Any Questions?

QUIZ!

Question

Which of these is <u>not</u> a method of data collection?

A) Questionnaires

B) Interviews

C) Experiments

D) Observations

Answer

Which of these is not a method of data collection?

A) Questionnaires

B) Interviews

C) Experiments

D) Observations